

Chromosphere

The Sun is surrounded by layers of gas, forming an atmosphere. The inner layer is called the chromosphere. The outer layer is called the corona, and extends into space for millions of kilometers.

Spicules

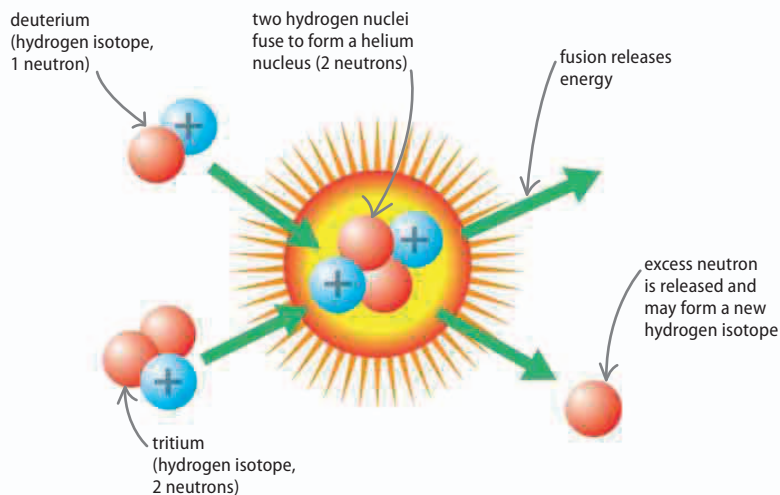
Jets of gas called spicules shoot up to 10,000 km (6,200 miles) from the photosphere for bursts of up to 10 minutes.

Sunspots

These dark areas are around 1,500°C (2,700°F) cooler than the rest of the surface. They occur where magnetism prevents hot gas from reaching the surface.

Nuclear fusion

Nuclear fusion occurs when the nuclei of two atoms fuse (join) together to make a large nucleus and energy is released. Every element is made up of atoms with a different number of protons (positive particles) in the nucleus and neutrons (no charge). Hydrogen, the most common element in the Sun, has one proton and usually no neutrons. However, the heat and pressure in the Sun's core increases the chance for hydrogen isotopes to form, with one or two neutrons. They fuse to form helium.



The **sunspots** on the Sun's surface may last from a few hours to **several weeks**.

△ Activity in the Sun's core

At the Sun's core hydrogen nuclei collide at great speed. The fusion process is complex, but one of the reactions that takes place is shown above. Here, two different hydrogen isotopes (see page 169) fuse to form helium, releasing energy and an excess neutron.

REAL WORLD**Little Ice Age**

The number of sunspots rises and falls over an 11-year cycle. It is believed that sunspot activity may affect the climate on Earth. In the late 1600s, there was a long period when few sunspots were recorded, which coincided with a series of very cold winters in Europe that became known as the Little Ice Age. For about 100 years the Thames River, in London, England, froze almost every winter, and frost fairs were held on the thick ice.